

SATHWIK H NAIK

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PROFESSIONAL SUMMARY

Recent M.S. Data Science graduate (University of Maryland, College Park, 2026, GPA 3.81/4.0) with ~1 year of corporate analytics experience at a Fortune 50 healthcare organization and eight substantial technical projects spanning data engineering, streaming reliability, BI/analytics, AI/RAG engineering, and data science with causal inference. This master document consolidates full project and skills depth across all target identities (Data Analyst/BI, Data Scientist, Data Engineer, Analytics Engineer, AI/ML Engineer); individual applications use a tailored, identity-locked, one-page version drawing from this reference.

EDUCATION

University of Maryland, College Park

M.S. in Data Science

2024 – 2026

GPA 3.81/4.0

Coursework Statistical Inference, Deep Learning, Data Modeling, Data Visualization, Experimental Design, Advanced Time Series

Reva University, India

B.Tech, Computer Science and Engineering

2019 – 2023

GPA 3.55/4.0

WORK EXPERIENCE

Elevance Health

Aug 2023 – Aug 2024

Software Engineer — Fortune 50 healthcare organization

Tech Stack: Salesforce CRM, Informatica, Bitbucket, Power BI, DAX, AutoRABIT, Apex

- Designed reusable KPI structures, translating operational requirements into consistent, self-serve cross-team metrics
- Built and maintained 10+ interactive Power BI dashboards with DAX measures for multiple internal teams
- Developed ETL workflows to source, clean, and validate healthcare data, improving quality for stakeholder reports
- Automated recurring data-preparation and reporting workflows, eliminating 20+ hours of manual weekly effort
- Documented report logic, metric definitions, and validation rules enabling governed self-service reporting
- Supported CI/CD pipelines for Salesforce releases using AutoRABIT with automated logging and standardized workflows

PROJECTS

Healthcare Insurance Claims Analytics & Fraud Signal Platform ([GitHub](#))

Tech Stack: Apache Kafka (Redpanda), Apache Spark (batch + Structured Streaming), Delta Lake/Databricks (Unity Catalog), dbt, Apache Airflow, Snowflake, Terraform, Python, Docker Compose

- Built a hybrid lakehouse-warehouse pipeline using Kafka, Spark, Delta Lake, dbt, Airflow, and Snowflake
- Implemented Bronze/Silver/Gold medallion architecture on Delta Lake processing 50,000 synthetic insurance claims
- Built two dbt projects with tests, transforming raw claims into governed, BI-ready Snowflake marts
- Orchestrated the end-to-end pipeline in Airflow with retries, exponential backoff, and failure alerting
- Provisioned Databricks and Snowflake infrastructure declaratively using Terraform with key-pair JWT authentication
- Built a rolling-baseline fraud detection rule in dbt achieving 100% recall and 67% precision
- Resolved a live Kafka at-least-once duplicate-delivery event using existing Silver-layer deduplication logic
- Built Spark Structured Streaming ingestion using explicit schemas, checkpointing, and micro-batch triggers

AI Support Telemetry Reliability Platform ([GitHub](#))

Tech Stack: TypeScript, Node.js 22, Amazon Kinesis (via LocalStack), Amazon S3, AWS CDK, AWS Lambda, Ajv (JSON Schema), DuckDB, npm workspaces, Vitest, Docker

- Built a TypeScript streaming telemetry platform using Kinesis-compatible APIs, LocalStack, S3, DuckDB, and AWS CDK
- Reconciled 32,154 synthetic transport records to zero unaccounted loss across accepted and quarantine zones
- Designed at-least-once canonicalization removing 600 injected duplicates with zero duplicate canonical keys
- Enforced versioned JSON Schema contracts via Ajv, isolating 50/50 invalid records at 100% precision and recall
- Ran seven deterministic reliability experiments covering burst load, outage recovery, and contract evolution
- Drained a backlog from a 60-second simulated consumer outage in 1.748 seconds after resuming
- Synthesized a least-privilege AWS CDK stack with Lambda partial-batch handling and CloudWatch alarms
- Modeled synthetic AI-model cost, latency, and resolution trade-offs in DuckDB with Wilson confidence intervals

SaaS Subscription Churn & Revenue Analytics Platform ([GitHub](#))

Tech Stack: Python (pandas, NumPy), DuckDB (SQL), PostgreSQL, Grafana (config-as-code), Plotly, scikit-learn (Gradient Boosting), SHAP, Docker Compose

- Built a SaaS churn analytics platform across DuckDB, pandas, PostgreSQL, Grafana, and scikit-learn
- Modeled the MRR waterfall and running balance in SQL using CTEs and window functions
- Computed cohort retention matrices using CROSS JOIN and FILTER-based conditional aggregation in SQL
- Reconstructed point-in-time customer state using pandas merge_asof across 8,000 synthetic customer accounts
- Derived 15+ metrics tables including NRR, GRR, and ARPA from raw subscription events
- Built a 21-panel Grafana dashboard provisioned entirely as version-controlled config, not UI clicks
- Designed a four-tier dashboard hierarchy with live template-variable filters for plan and channel
- Trained a Gradient Boosting churn model and explained predictions per-customer using SHAP

RAG Evaluation Lab ("Retrieval Quality Lab") [\(GitHub\)](#)

Tech Stack: Python, Hugging Face datasets (SQuAD), rank_bm25, sentence-transformers (BGE-small, cross-encoder reranker), ChromaDB (HNSW), Ollama, RAGAS, LangChain, Streamlit, Plotly

- Built a RAG evaluation pipeline comparing BM25, dense, hybrid, and reranked retrieval methods
- Achieved 85.3% Recall@1 with hybrid retrieval plus cross-encoder reranking, up from 69.6% BM25
- Evaluated generation quality using EM, F1, a custom grounding score, and RAGAS relevancy
- Diagnosed generation, not retrieval, as the system bottleneck despite 95% gold-doc accuracy
- Built a failure taxonomy classifying generation errors across five distinct, prioritized failure modes
- Built a 5-tab Streamlit dashboard visualizing retrieval ablations, generation quality, and latency
- Ran a local Ollama LLM for generation and LLM-as-judge evaluation with no paid APIs
- Indexed 9,000 SQuAD documents using BM25 and Chroma dense vector search with HNSW

Banking Liquidity Forecast Lab [\(GitHub\)](#)

Tech Stack: Python, LightGBM, statsmodels (Holt-Winters, OLS with HC1 robust standard errors), FastAPI, Docker Compose

- Built a retail-banking cash-outflow forecasting pipeline comparing seasonal naive, Holt-Winters, and LightGBM
- Achieved 9.48% WAPE forecasting daily cash outflow with Holt-Winters under six-fold walk-forward validation
- Engineered leak-safe lag, rolling-window, and Fourier calendar features for multi-segment LightGBM forecasting
- Built a residual-based anomaly detector using robust z-scores, reaching 0.71 F1 at threshold
- Swept anomaly thresholds from 2.0 to 5.0, mapping precision-recall tradeoffs to operations cost
- Measured a marketing campaign's causal lift using interrupted time series and difference-in-differences
- Recovered a known 12% campaign lift within 0.4 percentage points using two causal estimators
- Served the forecasting model through FastAPI endpoints packaged in reproducible Docker Compose containers

Conditional Generative Data Augmentation for Image Classification [\(GitHub\)](#)

Tech Stack: Python, PyTorch, Conditional GAN (cGAN), Conditional VAE-GAN, CNN classifier, MNIST, Fashion-MNIST

- Developed CGAN and Conditional VAE-GAN pipelines in PyTorch across MNIST and Fashion-MNIST
- Improved Fashion-MNIST accuracy from 63.6% to 70.8% using only 100 real samples
- Evaluated seven classifier training strategies, finding weighted real-synthetic mixing performed best overall
- Ran architecture and training-strategy ablations, finding learning-rate ratio the strongest performance lever
- Compared cGAN against a Conditional VAE-GAN hybrid across two datasets and twelve experiments
- Found VAE-GAN augmentation underperformed real-only training, showing complexity alone doesn't improve results
- Identified a large-latent VAE-GAN variant as the only architecture improving both dataset's F1

TinyACE: Bounded Working-Memory Context Engineering for Small Language Models [\(GitHub\)](#)

Tech Stack: Python, PyTorch, TinyLlama-1.1B, Phi-3 Mini (3.8B), Mistral-7B, sentence-transformers (SBERT), NVIDIA A100

- Developed TINYACE, a bounded working-memory framework capping context at 512 tokens for small models
- Identified a Capacity Sweet Spot where mid-sized Phi-3 Mini gained 4 accuracy points
- Improved Phi-3 Mini accuracy from 74% to 78% across a 50-question SciQ benchmark
- Ran a multi-scale evaluation across TinyLlama, Phi-3 Mini, and Mistral on one GPU
- Identified failure tracking as the most critical component through systematic ablation studies
- Found simple FIFO eviction matched or beat complex utility scoring for small models
- Designed a vagueness-detection gatekeeper preventing generic entries from bloating the memory playbook

Advanced Time Series Analysis & Predictive Dashboarding

Tech Stack: Python, Docker, Git, statistical trend analysis, rolling time-window ingestion, anomaly detection

- Built a refreshable ingestion pipeline monitoring rolling 24-hour windows of operational metrics
- Applied moving averages and trend signals to detect anomalies in high-frequency data
- Dockerized the workflow and used Git version control for reproducible execution across environments

TECHNICAL SKILLS

- **Analytics & BI** Power BI, DAX, Tableau, Grafana, Plotly, Streamlit, Data Visualization
- **Data & SQL** SQL, MySQL, ETL, Data Pipelines, Data Cleaning, Snowflake, DuckDB, PostgreSQL, Informatica, Data Governance
- **Data Engineering & Orchestration** dbt, Apache Kafka (Redpanda), Apache Spark (incl. Structured Streaming), Apache Airflow, Terraform, Databricks (Delta Lake, Unity Catalog)
- **Streaming Data Reliability & Cloud (AWS)** TypeScript, Node.js, Amazon Kinesis, Amazon S3, AWS CDK, AWS Lambda, Avro/JSON Schema, Vitest, Data Contracts, Idempotent Canonicalization
- **LLM & RAG Engineering** RAG Architecture, Hybrid Search (BM25 + Dense + RRF), Cross-Encoder Reranking, ChromaDB, Ollama, RAGAS, LangChain, sentence-transformers, Hugging Face Datasets
- **Statistics & Methods** Statistical Inference, Experimental Design, A/B Testing, Time-Series Forecasting, Causal Inference (Interrupted Time Series, Difference-in-Differences), Walk-Forward Validation, Model Evaluation
- **ML & AI** PyTorch, scikit-learn, Gradient Boosting, LightGBM, SHAP, GANs, VAE-GANs, CNNs, Deep Learning, LLM Context Engineering
- **Programming & Tools** Python, pandas, NumPy, PyArrow, R, Git, Bitbucket, Docker, Docker Compose, FastAPI, Jupyter, CI/CD
- **Salesforce Ecosystem** Salesforce CRM, Apex, AutoRABIT
- **AI-Assisted Development** GitHub Copilot, Claude AI, Cursor, OpenAI Codex, AI-Assisted Debugging